Cellular Neuroscience: Neurons, Glia, Myelin, and Neural Tissue Responses

Chapter 4

In this chapter

Learning objectives 65
Introduction 65
4.1 Cell theory and nervous tissue 65
4.2 Neuronal structure 66
4.3 Membrane potentials 66
4.4 Action potentials 67
4.5 Astrocytes 67
4.6 Microglial surveillance 68
4.7 Oligodendrocytes and myelin 69
4.8 Blood–brain barrier 69
4.9 Cellular stress responses 70
4.10 Cellular repair and adaptation 70
Synthesis 71
Composite case study 71
Review and discussion questions 71
Research assignments 72
Chapter summary 72
References 72

Central question. What happens inside and between cells when the nervous system responds to threat, and which of those events can be inferred from human evidence?

Learning objectives

- describe neuronal structure and the ionic basis of membrane and action potentials;
- explain the roles of astrocytes, microglia, oligodendrocytes and the blood–brain barrier;
- distinguish acute cellular stress responses from lasting injury;
- evaluate claims that trauma "damages" or "rewires" cells;
- relate cellular evidence to clinical and research questions.

Introduction

A textbook of trauma neurobiology can be written entirely at the level of circuits, but circuits are made of cells, and cells are where the mechanisms ultimately reside. Synaptic change, hormonal action, immune signalling and gene expression all converge on neurons and glia. Understanding this level prevents two opposite errors: treating the brain as an undifferentiated blob that "lights up", and treating molecular findings as though they explained a person's life.

This chapter describes the cellular machinery that matters most for traumatic stress: how neurons generate and transmit signals, how glia support and modulate them, how the blood–brain barrier regulates what reaches the tissue, and how cells respond to stress. It closes with the interpretive problem that dominates this literature: most of the strongest cellular evidence comes from animals, and the gap between a dendritic spine in a rodent and a symptom in a patient is wide.

> **A note on scale.** *A typical human brain contains roughly 86 billion neurons and a comparable number of glial cells, connected by an estimated 100 trillion synapses. No technique currently measures all of this in a living person. Every human method samples a proxy: blood flow, electrical fields, tissue volume, or post-mortem structure.*

4.1 Cell theory and nervous tissue

The cell theory holds that all living organisms are composed of cells, and that cells arise from existing cells. Its application to the brain was

delayed because neurons have extraordinarily complex shapes and were difficult to resolve with early microscopes. The work of Ramón y Cajal and others established the neuron doctrine: nerve cells are individual units that communicate at contacts rather than forming a continuous network.

Nervous tissue contains neurons and glia. Neurons are electrically excitable and transmit information; glia were long described as support cells but are now recognised as active participants in signalling, immune surveillance, myelination and the regulation of the extracellular environment. Nervous tissue also contains blood vessels, whose endothelial cells form the blood–brain barrier, and extracellular matrix, which influences how cells migrate and connect.

The cellular account given here follows standard neuroscience teaching; its stress-specific implications are summarised by Godoy et al. (2018).

4.2 Neuronal structure

A neuron has a cell body (soma), dendrites, an axon and presynaptic terminals. Dendrites receive input; the axon carries output; terminals release neurotransmitter onto the next cell. The axon hillock is where action potentials are typically initiated; the axon may be myelinated, which increases conduction velocity. Neurons are classified by morphology (pyramidal, stellate, bipolar), by neurotransmitter (glutamatergic, GABAergic, dopaminergic), and by projection pattern (local interneuron versus long-range projection neuron).

Cortical pyramidal cells are the principal output neurons of the cortex. Their apical dendrites extend toward the cortical surface, where they receive input from other cortical areas and from the thalamus. Their structure is plastic: dendritic spines, the small protrusions that host most excitatory synapses, change shape and number with learning and with stress. This is the structural basis of the claim that experience "shapes" the brain—though the changes are typically modest, reversible and regionally specific.

4.3 Membrane potentials

Neurons maintain a resting membrane potential of about −70 millivolts, inside relative to outside. The potential arises from selective permeability and from the unequal distribution of ions, principally sodium, potassium, chloride and calcium, maintained by pumps such as the sodium–potassium ATPase. At rest, the membrane is more

permeable to potassium than to sodium, so the resting potential sits near the potassium equilibrium potential.

Changes in permeability produce graded potentials. Excitatory inputs open cation channels and depolarise the membrane; inhibitory inputs open chloride or potassium channels and hyperpolarise it. These inputs sum over space and time at the axon hillock. If the summed depolarisation reaches threshold—around −55 millivolts—voltage-gated sodium channels open and an action potential is initiated.

This machinery matters clinically because many drugs act on it. Benzodiazepines enhance GABA-A receptor function; anticonvulsants block sodium channels; anaesthetics modulate ion channels and synaptic release. Trauma-related conditions are frequently treated with medications that act at this level, and understanding it clarifies both effects and adverse effects.

4.4 Action potentials

An action potential is a rapid, all-or-none change in membrane potential. Voltage-gated sodium channels open, sodium enters, and the membrane depolarises to about +40 millivolts. Sodium channels then inactivate, potassium channels open, and the membrane repolarises, often overshooting into a brief afterhyperpolarisation. The refractory period that follows limits firing frequency and ensures that the signal travels in one direction.

Action potentials propagate along the axon. In unmyelinated axons, propagation is continuous and relatively slow. In myelinated axons, the signal jumps between nodes of Ranvier—saltatory conduction—which is faster and more energy-efficient. Myelin therefore does not merely insulate; it determines the timing of signals, and timing is functionally important across distributed circuits.

Action potentials are not the whole story of neuronal communication. Many neurons also fire in rhythmic bursts, and subthreshold oscillations coordinate activity across regions. The gamma oscillations implicated in attention and memory, for example, depend on the precise interaction of excitatory and inhibitory cells.

4.5 Astrocytes

Astrocytes are the most abundant glial cells in the central nervous system. They extend processes that contact synapses, blood vessels and

other astrocytes. They regulate the extracellular concentration of ions and neurotransmitters, particularly glutamate, which they take up and convert to glutamine. They contribute to the blood–brain barrier, to metabolic support, and to synapse formation and elimination.

Astrocytes also participate in signalling. They respond to neurotransmitters with calcium transients and can release gliotransmitters that modulate neuronal excitability. The idea that astrocytes form a "tripartite synapse" alongside pre- and postsynaptic elements is influential, though its functional importance in humans is still being established.

In stress and inflammation, astrocytes become reactive: they change morphology, alter glutamate uptake, and release cytokines. This may be protective in the short term and harmful if sustained, because excess glutamate in the extracellular space can be excitotoxic. The balance between astrocytic protection and astrocytic contribution to injury is an active research question, not a settled one.

4.6 Microglial surveillance

Microglia are the resident immune cells of the central nervous system. They derive from yolk-sac progenitors and populate the brain before birth. In the healthy brain they are highly motile, continuously extending and retracting processes to survey the parenchyma. They phagocytose debris, prune surplus synapses during development, and respond to injury and infection.

Activation is not a single state. Microglia adopt a range of phenotypes depending on context, and the older M1/M2 classification is now regarded as an oversimplification. In chronic stress and in some psychiatric conditions, microglial activation has been reported, but human evidence in vivo is limited and methods for measuring it are indirect.

The relevance to trauma is plausible but not established at the level often claimed. Stress can increase inflammatory signalling, and inflammation can affect mood, sleep and behaviour. Whether microglia mediate a specific PTSD symptom in humans remains unknown. Claims that trauma "inflames the brain" should therefore be translated into the more accurate statement that stress is associated with immune changes, some of which may affect the central nervous system.

Microglial involvement in stress-related conditions is reviewed by Hori and Kim (2019).

4.7 Oligodendrocytes and myelin

Oligodendrocytes produce myelin in the central nervous system. Each oligodendrocyte myelinates multiple axon segments, and the thickness and length of myelin sheaths affect conduction velocity. Myelination continues into adulthood in association cortex, and it is activity-dependent: electrically active axons are preferentially myelinated.

Myelin plasticity is a relatively recent addition to the plasticity literature, which for decades focused almost exclusively on synapses. It offers a mechanism by which experience could alter the timing and synchrony of distributed circuits. Stress and social isolation have been reported to affect myelination in animal models, and diffusion imaging in humans shows associations between early adversity and white-matter microstructure.

These findings are important but easy to overread. Diffusion metrics are indirect, affected by crossing fibres, partial volume and water content, and do not measure myelin directly. A group difference in fractional anisotropy is not evidence that a person's myelin is damaged.

4.8 Blood–brain barrier

The blood–brain barrier is formed by tight junctions between endothelial cells of cerebral capillaries, supported by pericytes and astrocytic endfeet. It restricts the passage of many molecules while allowing selective transport of glucose, amino acids and other essentials. It protects the brain from toxins and pathogens and maintains a stable ionic environment.

Circumventricular organs, such as the area postrema and the organum vasculosum of the lamina terminalis, lack a complete barrier and allow the brain to sample circulating signals. This is one route by which peripheral inflammation and hormones influence central function.

The barrier is dynamic, not absolute. Stress, infection and inflammation can increase permeability, and some evidence suggests altered barrier function in psychiatric and neurodegenerative conditions. In trauma research, the barrier matters conceptually because it explains how peripheral immune signals can reach the brain, and practically because it constrains which drugs can be used to treat central conditions.

4.9 Cellular stress responses

Cells respond to stress through conserved programmes. Heat-shock proteins help refold damaged proteins. The unfolded protein response adjusts protein synthesis in the endoplasmic reticulum. Autophagy removes damaged organelles. Antioxidant systems neutralise reactive oxygen species. When these systems are overwhelmed, cells may undergo apoptosis or necrosis.

Neurons are particularly vulnerable to energy failure and excitotoxicity because of their high metabolic demand and their dependence on tight ionic regulation. Ischaemia, hypoglycaemia, prolonged seizures and severe stress can all trigger cascades that begin with glutamate release and calcium influx and end with cell death.

In trauma, the relevant question is usually not whether cells die but whether they adapt. Glucocorticoids, catecholamines and inflammatory mediators alter gene expression, receptor density and dendritic structure. These adaptations can be protective—sharpening vigilance during danger—or costly when sustained. The distinction between adaptation and injury is central, and it is often blurred in popular accounts.

For a review of how stress mediators affect cellular and synaptic function, see Godoy et al. (2018) and McEwen (1998).

4.10 Cellular repair and adaptation

Repair in the central nervous system is limited but real. Axons in the peripheral nervous system can regenerate; those in the central nervous system generally cannot, largely because of inhibitory molecules in myelin and the glial scar. Synapses, however, are continually remodelled, and function can be restored through reorganisation as well as through repair.

Rehabilitation exploits this. Constraint-induced movement therapy, for example, improves function after stroke by forcing use of the affected limb, driving cortical reorganisation. Cognitive and behavioural therapies may work through analogous learning mechanisms, strengthening some circuits and weakening others.

The clinical implication is that the adult brain retains substantial capacity for change. The corollary is that change requires conditions: attention, repetition, safety and sleep. A cell that is chronically stressed, sleep-deprived and inflamed is not well placed to learn.

Synthesis

Table 4.1 summarises the principal cell types and their relevance to traumatic stress.

Table 4.1. Cell types and trauma-relevant functions

Cell type	Principal functions	Trauma-relevant relevance	Evidence limits
Neurons	Excitability, synaptic transmission, plasticity	Fear learning, extinction, regulation	Human measures are indirect
Astrocytes	Ion and glutamate homeostasis, barrier support, synapse regulation	Excitotoxicity risk, neuroimmune signalling	Mostly animal evidence
Microglia	Immune surveillance, synaptic pruning, phagocytosis	Inflammation, synaptic remodelling	Phenotypes oversimplified
Oligodendrocytes	Myelination, conduction timing	Circuit timing, white-matter findings	Diffusion measures indirect
Endothelial cells	Blood–brain barrier	Peripheral–central signalling	Human measurement invasive

Composite case study

A patient with chronic pain and PTSD reports that her symptoms worsen after infections. A cellular formulation would consider inflammatory signalling, sleep disruption, medication effects and pain sensitisation rather than asserting that infection "inflames her trauma". The practical response is to treat the infection, review sleep and analgesia, and monitor symptoms without attributing them to a single mechanism.

Review and discussion questions

Why is the resting membrane potential negative, and what determines the threshold for an action potential?

What does myelin do besides insulating axons?

How convincing is the evidence that microglia mediate trauma-related symptoms in humans?

Why are diffusion imaging findings often overinterpreted?

Distinguish cellular adaptation from cellular injury with an example of each.

What would a study need to show before claiming that a psychological intervention changed synaptic structure?

Research assignments

- Design an animal study of chronic stress and myelination, specifying controls for handling, housing and sex differences.
- Critically appraise one human study linking inflammation to PTSD, noting the limitations of peripheral markers.
- Write a plain-language explanation of the action potential for a non-specialist audience, with one accurate caveat about what it does not explain.

Chapter summary

Neurons generate and transmit electrical signals through ion channels and synaptic release. Glia regulate the environment, modulate signalling, myelinate axons and mount immune responses. The blood–brain barrier governs what reaches the tissue. Cells adapt to stress through changes in gene expression, receptor density and structure, and these adaptations can be protective or costly. Most detailed cellular evidence comes from animals, and the translation to human experience is indirect. Cellular knowledge should therefore inform mechanism and caution in equal measure.

References

Godoy, L. D., Rossignoli, M. T., Delfino-Pereira, P., Garcia-Cairasco, N., & de Lima Umeoka, E. H. (2018). A comprehensive overview on stress neurobiology: Basic concepts and clinical implications. *Frontiers in Behavioral Neuroscience, 12*, Article 127. https://doi.org/10.3389/fnbeh.2018.00127

Hori, H., & Kim, Y. (2019). Inflammation and post-traumatic stress disorder. *Psychiatry and Clinical Neurosciences, 73*(4), 143–153. https://doi.org/10.1111/pcn.12820

McEwen, B. S. (1998). Protective and damaging effects of stress mediators. *New England Journal of Medicine, 338*(3), 171–179. https://doi.org/10.1056/NEJM199801153380307